

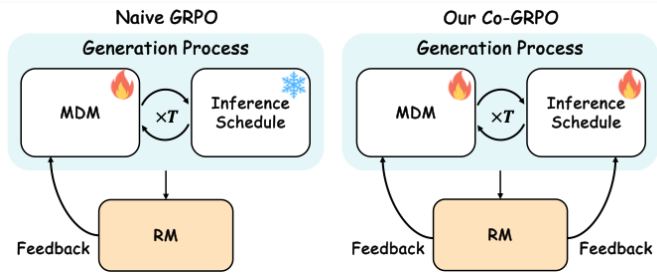


Co-GRPO: Co-Optimized Group Relative Policy Optimization for Masked Image Generation

Renping Zhou^{*1,2}, Zanlin Ni^{*1}, Tianyi Chen¹, Zeyu Liu¹, Yang Yue¹,
Yulin Wang¹, Yuxuan Wang¹, Jingshu Liu¹, Gao Huang^{1, ✉}
¹Tsinghua University ²Anyverse Dynamics



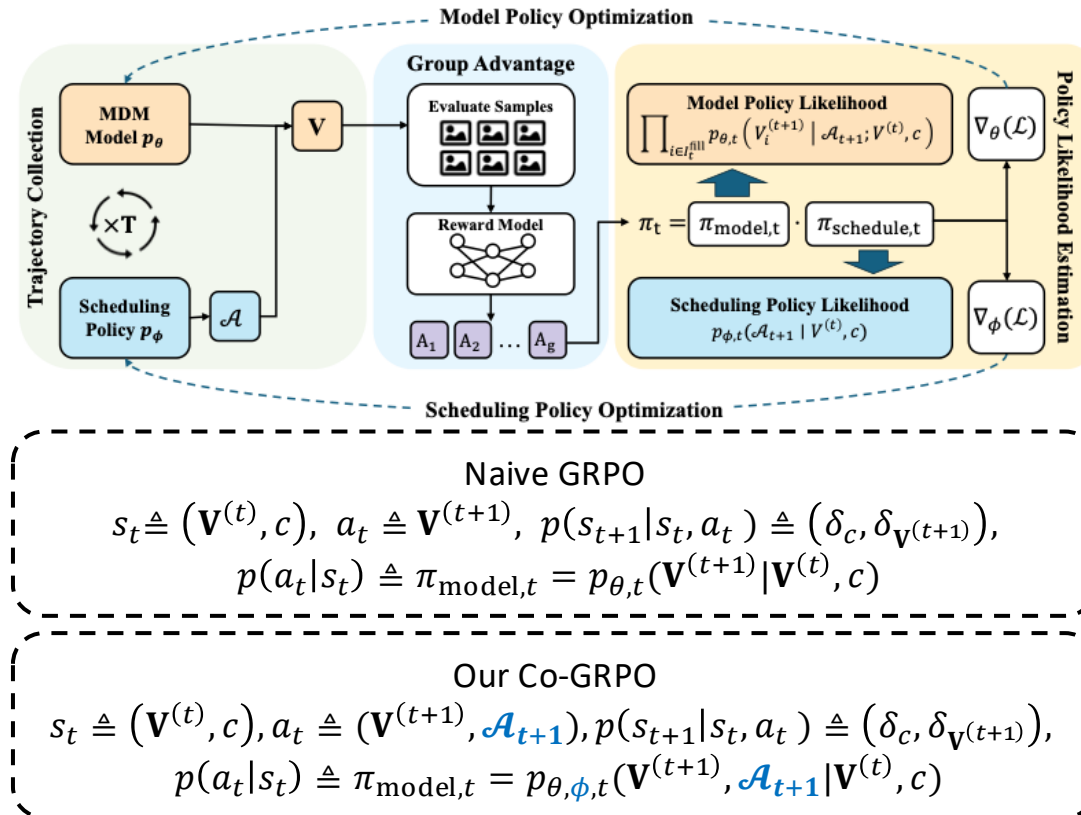
1. INTRODUCTION



Inference Schedule	Predefined Function
re-mask ratio $r(t)$	$r(t) = \cos(\frac{t+1}{T} \cdot \frac{\pi}{2})$
sample temperature $\tau_s(t)$	$\tau_s(t) = 1.0$
re-mask temperature $\tau_r(t)$	$\tau_r(t) = 2 \frac{T-t}{T}$
classifier-free guidance scale $s(t)$	$s(t) = 9.0$

- Co-GRPO jointly optimizes the MDM and inference schedule as cooperating policies in one MDP via trajectory-level policy gradients.
- Co-GRPO beats Naive GRPO on ImageReward / HPSv2 and zero-shot generalizes to GenEval / DPG-Bench.

2. APPROACH



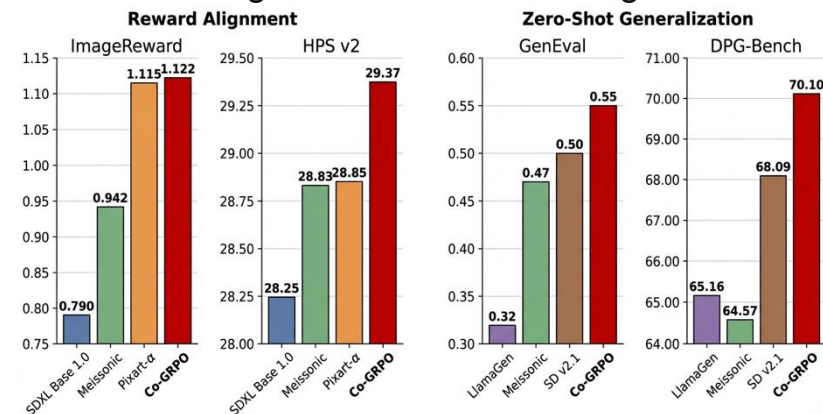
Steps	$\gamma = 1.0$	$\gamma = 1.5$	$\gamma = 2.0$
16	0.660	0.872	0.865
48	0.918	0.957	0.937

$r_t = \cos(\pi(t+1)/2T)^\gamma$
Scheduling critically affects MDM quality.

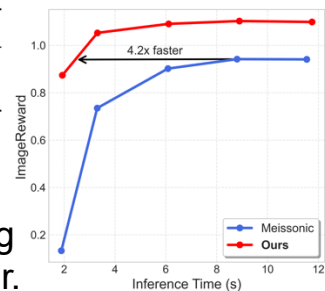
- **Joint MDP:** the agent acts on both visual tokens $\mathbf{V}$ and inference schedule $\mathcal{A}$.
- **Factorized Policy:** explicitly decouples the Model Policy π_θ from the Scheduling Policy π_ϕ .
- **Alternating Co-Optimization:** Instead of unstable joint updates, we iteratively optimize θ and ϕ to maximize the Group Relative Advantage.

3. EXPERIMENTS

- Co-GRPO consistently outperforms SOTA baselines across all metrics, excelling in both reward alignment and zero-shot generalization.



Method	ImageReward	HPSv2	GenEval
Show-o	1.028	28.53	0.64
+ Co-GRPO	1.216	29.17	0.75



- Co-GRPO demonstrates strong model-agnostic.
- Each step improves, hitting the same score 4.2x faster.

- Ablations demonstrates the effectiveness of our optimization strategy and its transfer capability.
- Qualitative comparisons shows that Co-GRPO produces images with finer details and better semantic alignment compared to the Base Model and Naive GRPO.

Strategy	Image.	HPS.
Base Model	0.942	28.83
Joint	1.031	29.06
Alternating	1.122	29.37

Method	Image.	HPS.	Cycle	Image.	HPS.
Base Model	0.942	28.83	0	0.942	28.83
+ Naive GRPO	1.014	28.86	1	1.067	29.21
+ Naive GRPO w/ S.S.	1.039	29.08	2	1.093	29.22
+ Co-GRPO	1.122	29.37	3	1.122	29.37

Steps	Test	Train	ImageReward			HPSv2		
			Baseline	Ours	Δ	Baseline	Ours	Δ
8	48		0.133	0.874	0.741	26.42	28.61	2.19
48	48		0.942	1.122	0.180	28.83	29.37	0.54
64	48		0.941	1.099	0.158	28.80	29.39	0.59



4. CONCLUSION

- We propose Co-GRPO, enhancing text-to-image generation in Masked Diffusion Models by incorporating the inference schedule into the underlying MDP.
- We develop a co-optimization strategy that jointly updates the inference schedule and model parameters, and validate it on Meissonic and Show-o, achieving significant improvements and strong generalization.

